

Mandy Xue

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EDUCATION

Santa Clara University, *M.Sc. Business Analytics* (GPA: 3.56/4.0) Sep. 2024 - Expected Dec. 2025

Relevant Courses: Machine Learning, Natural Language Processing, Large Language Model Applications, Database Management Systems, Prescriptive Analytics, Marketing Analytics, Text Mining

Hubei University of Education, *B.Sc. Applied Statistics* (Rank: 6/44, Score: 84.6/100) Sep. 2020 - June. 2024

Relevant Courses: Multivariate Statistical Analysis, Time Series Analysis, Optimization Modeling, Big Data Analytics

PUBLICATION

Xue, C. (2024, Apr). *Research on the Influencing Factors of Automobile Price Based on Multiple Linear Regression Model*.

In Proceedings of the 2nd International Conference on Green Transport, Renewable Energy and Environment (GTREE 2023). *Highlights in Science, Engineering and Technology*, Vol. 83. <https://doi.org/10.54097/w0tdrk16>

RESEARCH EXPERIENCE

Higher Education Address Database Cleaning Aug. 2025 - Sep. 2025

Research Assistant, Renmin University of China & Xi'an Jiaotong University

- Developed and implemented a **rule-based standardization framework** for institutional address data, employing regex include/exclude patterns to systematically resolve cross-linguistic inconsistencies, institutional name variants, and historical merger records.
- Automated the cleaning and integration of **over 1 million institutional records** using R and Python pipelines, which substantially improved bibliometric retrieval precision (from ~70% to >90%) and reduced noise in large-scale data collection.
- Generated a **validated benchmark dataset** that enables reproducible, China-focused bibliometric studies and facilitates comparative higher education research by improving the reliability of institutional affiliation matching across domestic and international datasets.
- Positioned this work as a methodological contribution to **computational social science and scientometrics**, demonstrating how advanced data engineering techniques can enhance the validity of large-scale bibliometric analyses and policy-relevant higher education research.

IMPAQTO - Impact Measurement & Data Analysis July. 2025 - Sep. 2025

Research Intern, Santa Clara University | Advisor: Anthony Sampson & Veronica Taco

- Cleaned and systematized a decade of longitudinal programmatic data**, harmonizing heterogeneous sources and constructing a Theory of Change–Dashboard Mapping Table to explicitly trace causal linkages between resources, activities, and intended social outcomes.
- Conducted **computational text analysis on 5,000+ chatbot feedback entries** using Python (topic modeling, sentiment classification, and keyword clustering) to extract recurring themes in participant experiences, providing a qualitative layer to quantitative outcome metrics.
- Built a **multi-dimensional Tableau dashboard** integrating over 100 program indicators, combining descriptive statistics, interactive visualizations, and narrative-driven outcome stories to enable **triangulation across quantitative KPIs and qualitative evidence**.
- Drafted a preliminary **impact evaluation framework** that aligns ToC models with international standards in social impact assessment, offering actionable recommendations to strengthen monitoring, evaluation, and learning practices for community-based organizations.

DigitalOcean - AI-Powered Billing Comparison System for Cloud Cost Optimization Jan. 2025- June. 2025

Practicum Researcher, Santa Clara University | Advisor: Prof. Tao Li

- Designed and implemented an **OCR + Large Language Model (LLM)-based billing comparison system** to automatically parse unstructured AWS invoices and standardize pricing data for comparative analysis.

- Constructed **LangChain-based Retrieval-Augmented Generation (RAG) pipelines** with custom vector embeddings to enable dynamic competitor cost evaluation, achieving up to **62.9% verified cost reductions** when substituting DigitalOcean pricing models.
- Deployed an **interactive Streamlit interface** to facilitate real-time competitor benchmarking, enhancing the transparency of cloud pricing structures for both researchers and practitioners.
- Positioned the project as a case study in **applied AI for cloud economics**, demonstrating how hybrid NLP architectures (OCR + RAG) can support computationally rigorous decision-making in **cost optimization and platform competition research**.

AutoKaggle: Multi-Agent Workflow System for Automated Data Science

Research Intern(remote) | Advisor: Wei Liu

Jan. 2024 - Jun. 2024

- Designed a **modular multi-agent system** for automated data science pipelines (data cleaning, feature engineering, model training), integrating LangChain and Streamlit.
- Standardized workflow orchestration by migrating to LangChain architecture, enabling **reusable memory management and interface consistency** for future academic research and industrial deployment.

Urban Planning & Governance

Apr. 2024 - Sep. 2024

Research Assistant, Shenzhen University | Advisor: Prof. Jin Guo

- Contributed to a **book chapter manuscript**, synthesizing statistical datasets and urban governance literature to theorize on **divergent governance trajectories** between shrinking and expanding cities.
- Conducted **OLS regression and GIS-based heatmap analysis** on multi-dimensional urban datasets, identifying service imbalances and infrastructure stress points.
- Proposed **dual governance strategies** combining pre-planning with adaptive reuse of assets, emphasizing resilience in urban development policy.

Marketing Strategies in Live-Stream Sales and Consumer Behavior

Undergraduate thesis | Advisor: Prof. Fen Wang

Dec. 2023 - Apr. 2024

- Applied **Structural Equation Modeling (SEM)** on 500+ survey responses to examine how live-stream marketing strategies shape consumer trust and purchase intention.
- Identified **trust as a mediating variable**, with influencer content quality, interactivity, and incentives emerging as significant predictors.
- Proposed strategic recommendations for **platform governance and marketing optimization** in e-commerce.

Automobile Price Determinants: A Multiple Linear Regression Analysis

Research Seminar | Advisor: Prof. Patrick Rebeschini (Oxford University)

Jan. 2023 - Mar. 2023

- Built regression models to evaluate **pricing determinants in automobile markets**, applying residual diagnostics to correct heteroscedasticity.
- Delivered **data-driven insights for automotive pricing strategies**, demonstrating the real-world applicability of rigorous statistical modeling.

PROJECT EXPERIENCE

Bank Customer Churn Prediction

Winter 2025 | Advisor: Prof. Tao Li

- Designed a **machine learning pipeline** on 540K anonymized banking records with a 20% churn rate, constructing domain-specific behavioral and transactional features (e.g., tenure, credit utilization, product mix) to capture latent customer attrition patterns.
- Applied **Synthetic Minority Oversampling Technique (SMOTE)** to address class imbalance and ensure robust generalization across underrepresented churn cases.
- Benchmarked six algorithms (Logistic Regression, Random Forest, XGBoost, LightGBM, SVM, MLP) under cross-validation, systematically comparing precision-recall trade-offs.
- Fine-tuned a **Multilayer Perceptron (MLP)** using grid search and regularization strategies, achieving **73% recall and 76% accuracy**, significantly improving the identification of at-risk customers.
- Interpreted model outputs with **SHAP value analysis**, highlighting account tenure and transaction frequency as the strongest predictors of churn, and situating findings within the broader literature on **predictive analytics for financial services**.

- Positioned results as a potential contribution to **data-driven retention strategy research**, demonstrating how advanced ML coupled with interpretable AI methods can support both managerial decision-making and academic inquiry in customer analytics.

Logistics Performance Enhancement

Fall 2024 | Advisor: Prof. Wilson Lin

- Conducted statistical and ML analysis on **300K+ retail transactions**, uncovering demographic patterns in purchasing and satisfaction outcomes.
- Evaluated shipping strategies, demonstrating that **same-day delivery increased “Excellent” feedback by 4.4%**, offering managerial insights into service differentiation.

WORK EXPERIENCE

Data Analysis Intern

Huazhong Changjiang Optoelectronics CO., LTD, (On-site)

May. 2023 - Dec. 2023

- Built a **SQL/ETL-based KPI dashboard** in Tableau to automate reconciliation across supply chain processes, reducing reporting workload by 30%.
- Designed a **predictive risk-control model** for supplier delays and cost fluctuations, identifying 15 high-risk nodes and improving efficiency by 20%.
- Experience contributed to **applied research on supply chain analytics** and operational risk modeling.

Data Analysis Intern

Wuhan Changjiang Computing Technology CO., LTD, (On-site)

Sep. 2022 - Apr. 2023

- Constructed a **conversion funnel in Power BI** to monitor leads, improving high-intent conversion by 25% and reducing CAC by 12%.
- Applied SQL and regression modeling to quantify **multi-channel marketing ROI**, increasing LTV by 8%.
- Positioned this work as a **practical application of marketing analytics research**.

Extracurricular Research Competitions

Statistical Modeling Contest — Hubei Province, 2023

- Led a team developing **epidemic prevention models** using Grey Prediction Models (GM(1,1)) to forecast demographic and economic recovery trends.
- Produced policy-oriented recommendations, integrating statistical theory with **public health applications**.

‘Internet’ Innovation & Entrepreneurship Competition — Gold Medal, Hubei Province, 2022

- Conducted **time-series forecasting** for the development of Dendranthema vestitum as a natural targeted drug for liver cancer.
- Proposed a **multi-channel distribution strategy** informed by competitive analysis and compound annual growth rate projections.

HONORS & AWARDS

- Dean’s List, Santa Clara University (Spring 2025)
- Outstanding Graduate, Hubei University of Education (2024)
- Finalist Thesis Award (Live-Stream Marketing Impact Study, 2024)
- 9th National College Students Statistical Modeling Contest – Hubei Province Third Prize (2023)
- 8th Mathematical Modeling Competition – University Level Third Prize (2022)
- 8th ‘Internet’ Innovation & Entrepreneurship Competition – Hubei Province Gold Medal (2022)

SKILLS

- **Machine Learning & NLP:** RAG pipelines, OCR integration, sentiment analysis, MLP classifier, time series forecasting
- **Data Analysis & Modeling:** SEM, OLS, customer segmentation, pricing strategies, churn prediction
- **Tools & Visualization:** Python (Pandas, NumPy, Scikit-learn), R, SQL, Tableau, Power BI, ETL, Excel, GIS, Matplotlib, Seaborn
- **Soft Skills:** Research writing, leadership, cross-functional collaboration, analytical storytelling